Chainstack: Enabling developers to unleash the power of Web3 with cloud technology

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About Chainstack

Headquartered in Singapore, Chainstack connects developers to Web3 infrastructure to power application development across decentralized finance (DeFi), non-fungible tokens (NFTs) and other industries. The business helps organizations in more than 150 countries reduce the time to market, costs and risks of building and scaling decentralized applications.

Industries: Technology

Google Cloud results

- Achieves minimum 99.9% availability for reliability requirements
- Accelerates release cycles from once every two months to several times a day
- Enables the Chainstack team to focus on build and development rather than infrastructure administration

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Location: Singapore

SQL (https://cloud.google.com/sql/),
(https://cloud.google.com/kubernetes-engine/),
(https://cloud.google.com/products/operations)
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Google Cloud (https://cloud.google.com/)

App Engine (https://cloud.google.com/appengine/)

Logging (https://cloud.google.com/logging/)

Monitoring (https://cloud.google.com/monitoring/)

Cloud SQL (https://cloud.google.com/sql/)

Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

Operations (https://cloud.google.com/products/operations)

The web has come a long way from the basic read-only functionality of web 1.0. Today, Web3 builds on the creativity, conversation and commerce unleashed by web 2.0 to create a wealth of opportunities for businesses and service providers.

Based in Singapore and the United States, Chainstack connects developers to Web3 infrastructure to power innovation across fast-maturing sectors, including decentralized finance (DeFi), non-fungible tokens (NFTs), gaming and analytics.

"We reduce the time to market, costs and risks involved in creating and scaling decentralized applications," explains Eugene Aseev, Chief Technology Officer and Founder of Chainstack.

Managing thousands of nodes worldwide with Kubernetes

To support more than 60,000 developers and thousands of companies in more than 150 countries worldwide, Chainstack operates a distributed cloud infrastructure with strategic locations on the west and east coast of the United States, across Europe and in Asia.

The business builds data APIs, and thousands of multi-chain nodes across 20 blockchain networks, providing decentralized storage and services distributed across six hosting providers and 10 regions to meet builders' and developers' needs. These providers include Google Cloud, other multinational cloud services and bare-metal infrastructure providers.

To orchestrate systems, applications and services across this infrastructure, Chainstack turned to Kubernetes (https://cloud.google.com/learn/what-is-kubernetes), the container orchestration system developed initially by Google before being open-sourced in 2014

"We were on the way to thousands of nodes and turned to containerization, as using traditional virtual machines or servers would have made our management load next to impossible," explains Aseev.

"Kubernetes met our orchestration needs and enabled us to run the same components in our technology stack as two and a half years ago, despite our infrastructure being around 100 times bigger. We can scale our stack further, future-proofing it as we continue to grow," he adds.

Building a core platform that's reliable, scalable and easy to manage

To fulfill its mission, Chainstack needed to develop and run its core platform, including control plane, building services, orchestration and security systems, on a reliable, managed, scalable and cost-effective infrastructure. So the business turned to

Google Cloud as its engineers had established relationships with account and technical team members from the cloud services provider.

Through these relationships, the startup was well aware of Google Cloud's reliability and performance, and its track record of delivering products and services that help solve clients' problems.

Ultimately, the opportunities Google Cloud presented to the Chainstack engineering team to unleash the potential of Web3, while minimizing infrastructure administration, prompted the business to use the cloud service to develop its platform.

"We reviewed cloud providers and Google Cloud was the best fit for our technical requirements. They had a roadmap for blockchain and new web iterations that complemented our own, and supported our engineering strategy of using managed services to minimize our team's infrastructure administration load," says Aseev.

Google Kubernetes Engine (https://cloud.google.com/kubernetes-engine) (GKE) enabled Chainstack to build in a fully, managed, serverless environment, allowing it to take advantage of the product's scalability and performance.

Google Cloud's Operations Suite (https://cloud.google.com/products/operations), including Monitoring and Logging (https://cloud.google.com/products/operations) monitors performance and provides diagnostic data, while the business runs its PostgreSQL and MySQL databases in Cloud SQL (https://cloud.google.com/sql) to take advantage of features such as automated database provisioning and storage capacity management.

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—Eugene Aseev, Chief Technology Officer and Founder, Chainstack

Eliminating infrastructure issues and providing room for growth

Running its platform in Google Cloud has paid off for Chainstack, with the business not experiencing any serious infrastructure issues. "Google Cloud, in particular, GKE, enables us to hit key technical metrics for our core platform," says Aseev. "For example, our cloud infrastructure gives us near-perfect reliability, at least 99.9% uptime."

With the platform running smoothly, the business can focus on continuing to build a business that has grown to the point its distributed cloud infrastructure handles more than 30,000 requests per second. It has also increased the number of chains and application chains supported to 20, up from just three in 2019.

With Google Cloud, Chainstack has established an impressive record of fulfilling demanding requirements that enable clients to provide high quality marketplaces, applications and other Web3-based products that meet customer needs. For example, Newgem, an European marketplace for NFTs, emerging artists and creators, relies on the Chainstack platform and infrastructure to enable users to create, purchase, sell and auction their digital creations on the Ethereum, Polygon and Avalanche networks. This enables the artists and creators to take advantage of growing demand for NFTs to increase their sales and revenue.

In addition, GET Protocol, an NFT ticketing infrastructure provider, has eliminated disappearing transactions and fallback code implementations for non-propagated transactions by running on Chainstack infrastructure. GET Protocol also receives real-time data that notifies the company of completed transactions within milliseconds, enabling them to prepare for any dropped requests.

Another client, UniWhales, provides real time transaction data and alerts for Swaps, DEX Liquidity, NFTs, Bridges, and top wallets in Web3 and DeFi. The company needed a node infrastructure that accurately provides real-time transaction data across multiple chains, so it turned to Chainstack for consistent and dependable access to blockchain networks with a distributed network of elastic nodes across various deployment locations.

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Accelerating release cycles to multiple times per day to drive Web3 adoption

With growth comes demand from customers and internal teams to release new features and products, and to do so quickly. To meet this need, Chainstack combines Google Cloud with continuous integration/continuous development (CI/CD) and better engineering practices to expedite its software release cycles. From a starting point of one release every two months, Chainstack has accelerated to a few releases daily.

"Three years ago, we were a small but nimble team of 15 people. Now we have over 100 people and we ship by order of magnitude faster than we used to," shares Aseev. "Google Cloud provided the tools that enabled us to start quickly and continued to work perfectly for us as we scaled."

With about four years of experience with Web3, Cloudstack also sees synergies emerging with Google Cloud, as the provider positions itself as a leader in the evolving web.

"We are transitioning Chainstack to an ecosystem of services that connects Web3 developers to decentralized infrastructure," explains Aseev. "Google Cloud enables us to provide a fast and reliable infrastructure and deliver globally distributed products and services."

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